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Prevalence of Hepatitis B virus and its predictors among volunteer blood donors in Jimma, Ethiopia, 2018: A cross-sectional study

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ABSTRACT

Background: Previous studies of blood donors in Ethiopia have focused on all types of donors without exposure screening before blood donation. The purpose of this study was to determine the prevalence of Hepatitis B virus and its predictors among volunteer blood donors in Jimma, Ethiopia.

Materials and methods: Cross-sectional study was conducted on sampled volunteer blood donors who were consecutively included from March 10 to April 20, 2018. Virus detection was investigated by testing for Hepatitis B surface antigen in the serum. Data collected through face-to-face interviews, has been cleaned and checked, entered into Data 3.1 and analyzed by statistical software SPSS version 20. The level of statistical significance was reported to be $p<0.05$ in the multivariable logistic regression.

Results: A total of 548 participants with 60.96% in the 18–24 age group were participated and the overall prevalence of HBV infection was 2.92%. The test positivity rate among males was 12/268 (4.48%) and while the rate among females was 4/280 (1.43%). More than 80% of those who tested positive were under the age of 35 years. Being male [AOR=3.28, 95%CI: 1.01–10.68], age 18–24 [AOR=0.17, 95%CI: 0.36–0.78], frequency of donation [AOR= 0.25, 95%CI: 0.08–0.76], and exposure to unsafe injection [AOR= 6.98, 95%CI: 1.66–29.29] were significant factors.

Conclusion: The overall prevalence of HBsAg was intermediate with positivity higher in males. Furthermore, age, frequency of donation and exposure to unsafe therapeutic drug injection were independent predictors. Therefore, the blood bank should raise awareness to repeat volunteer young donors and focus on identified low-risk groups.

1. Introduction

Hepatitis B is a potentially life-threatening liver infection caused by Hepatitis B virus (HBV) and it is a major global health problem [1]. It can cause both acute and chronic disease [2,3]. Hepatitis B is a silent killer liver disease for which many carriers are unaware of its clinical status; therefore, they act as a potential source of infection for other seronegative people [4]. The presence of Hepatitis B surface antigen (HBsAg) in the blood indicates that a person has been infected with the virus. Individuals who recover from acute hepatitis B infections clear the blood off HBsAg within approximately four months after the onset of symptoms. These individuals develop antibodies to HBsAg (anti-HBs) that provides complete immunity to subsequent hepatitis B viral infection. Similarly, individuals who are successfully vaccinated against hepatitis produce anti-HBs in the blood [5].

The main routes of HBV transmission are by percutaneous or mucosal exposure to infected blood and various body fluids, as well as through saliva, tears, menstrual, vaginal, and seminal fluids. Transmission of the virus may also occur through the reuse of needles and syringes either in healthcare settings or among persons who inject drugs. Also, the infection can occur during medical, surgical, and dental procedures, through tattooing, or through the use of razors and similar objects that are contaminated with infected blood [6].

Worldwide, it is estimated that 240 million people are chronically infected with HBV [7]. According to the latest reports, 68,600 people worldwide die each year from HBV infection and more than 300,000 from liver cancer as a result of hepatitis B [8]. The prevalence of HBV infection is classified as high when it is above 8%, as intermediate when it is between 2% and 7%, and as low endemicity when it is below 2% [9]. The prevalence of HBV is estimated to be ≥8% in West Africa and

Abbreviations: AOR, Adjusted Odds Ratio; CI, Confidence Interval; COR, Crude Odds Ratio; ELISA, Enzyme Linked Immune Sorbent Assay; HBsAg, Hepatitis B Surface Antigen; HBV, Hepatitis B Virus; HCV, Hepatitis C Virus; HIV, Human Immunodeficiency Virus; PCR, Polymerase Chain Reaction; SD, Standard Deviation; WHO, World Health Organization.

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5–7% in East Africa [10]. Ethiopia, being part of East Africa region, is classified as an area with a medium HBV endemic rate. The prevalence of HBV infections in blood donors in various parts of the world is between 0.008% and 6.08% [11]. Among voluntary and unpaid blood donors recruited through effective education and screening programs, the prevalence and incidence of HBV infection should be lower than in the general population [11,12].

Contaminated blood transfusions cause up to 16 million new HBV infections worldwide. For every unit of blood transfused, there is always a 1% probability of transfusion-related risks, including transfusion-related infections [13,14]. Worldwide, an estimated 1.6 million blood units are discarded annually due to the presence of markers for transfusion-borne infections such as Human Immunodeficiency (HIV), HBV, Hepatitis C Virus (HCV), and syphilis [11].

In Sub-Saharan Africa, 12.5% of patients who have received a blood transfusion are at risk of post-transfusion hepatitis [15]. It is estimated that more than 45,000 HBV infections are transmitted by contaminated transfusions each year [16]. From 2000 to 2011, the number of Sub-Saharan African countries testing at least 95% of donated blood units for HBV increased from 76% to 94% and the HBV marker-reactive units decreased from 7.1% to 4.4% [17].

Several programs have been implemented in Ethiopia to reduce the exposure to HBV in the population which include the expanded program on immunization policy, including vaccinating children against HBV in the form of pentavalent at 6, 10, and 14 weeks of age after birth; HBsAg screening of all pregnant women during the prenatal phase and vaccination of their babies at birth, as well as recommending the vaccination of high-risk groups, such as health professionals, against HBV. However, these programs have not been routinely implemented in most healthcare facilities across the country [18].

The World Health Organization (WHO) recommends quality-assured screening of all blood donations for transfusion-borne infections, including HIV, HBV, and HCV, and has been adopted by the Ethiopian government for the availability of safe and effective blood and blood components [6]. Previous studies of the prevalence of HBV infection and its predictors in blood donors in Ethiopia have focused on all types of donors, including familial replacement blood donors and commercial blood donors, where the prevalence of HBV is high due to the lack of exposure testing prior to donating blood.

Therefore, this study was aimed to determine the prevalence of HBV and related factors in voluntary blood donors from the Jimma Zone, which will help prevent and control HBV infection and ensure blood safety in pursuing national blood transfusion services guaranteed, strategies and improvement of the blood donor selection protocol by geographic area and population subgroup.

2. Materials and methods

2.1. Study setting and period

The study was carried out in Jimma Zone ofOromia Regional State, which is 354 km from Addis Ababa, the capital city of Ethiopia. According to 2017 data from the Jimma Zonal Health Office, the zone has an estimated 3174,484 total population. It has 21 districts with a total of 555 villages. There is a branch of the National Blood Transfusion Service Agency in the area that provides blood donation services to hospitals. A total of 678 units of blood donations were collected during the study period [19]. The data were collected from March 10 to April 20, 2018.

2.2. Study design and participants

A cross-sectional study design was used which the study participants were voluntary blood donors who visited the Jimma blood bank center and blood collection sites during the study period, lived in the area for the past six months, and were eligible for blood donation according to the national blood bank guideline for blood donation eligibility criteria which was adapted from the WHO blood donor selection guidelines on assessing donor suitability for blood donation [6] and includes:

Age: 18–65 years old, Weight: 45 kg and above
The interval between blood donations should not less than 3 months
Hemoglobin level within the normal range: for male ≥13.5 mg/dl; for female ≥12 mg/dl
Normal blood pressure: systolic/diastolic 100/70 up to 150/90 mmHg
No previous history of hepatitis, not having venereal diseases, and bleeding tendency
No chronic skin disease, heart disease, kidney disease, and diabetes mellitus
Not in a condition of acute weight loss up to 10% within last six months
Not be a homosexual
No history of drug addiction by injection
Not less than 6 months after operation or delivery (termination of pregnancy), and
Not be pregnant or lactating women.

Blood donors who refused to give their consent were excluded.

2.3. Study variables

Dependent variable: HBV infection status
The independent variables were;

Sociodemographic factors: age, sex, marital status, educational level, place of residence, occupation
Behavioral factors: sharing razors and sharp objects, multiple unprotected sexual activities, tattoos, ear/nose piercings, injection of unsafe therapeutic drugs, number of blood donations
Knowledge-related factors: Knowledge of pathogens, routes of transmission, signs and symptoms, and treatment of HBV
Clinical/treatment-related factors: Blood transfusion, surgical and dental procedures, vaccination status, medical history

2.4. Sample size determination

The sample size was calculated manually using a single population proportion formula taking into account the prevalence of HBV among blood donors in southwestern Ethiopia 3.05% [20], margin of error 1.5%, and CI 95%. Assuming a 10% non-response rate, the final sample was 555. The prevalence of HBV among blood donors gives the sample size larger than the associated factors and was therefore used to determine the final sample size.

2.5. Sampling techniques

All voluntary blood donors were sequentially selected from the collection points in the Jimma Zone and the Blood Bank Center during the study period until the required sample was obtained. Three functional teams from blood bank staff were formed to collect blood from volunteer blood donors. Each team planned to collect 250 units of blood in ten days at the blood bank center and fifteen days outreach, depending on the number of volunteers at each planned collection site. Therefore, the sample size was evenly distributed among the case teams.

2.6. Data collection tools and techniques

The data were collected using a structured questionnaire that was adapted from the previous literature [8,24]. The questionnaire was originally prepared in English and translated into the local languages, Afan Oromo and Amharic for fieldwork purposes and back into English in order to check the language consistency.

After the donors agreed to participate in the study, they signed an informed consent form and the data collectors collected basic information from each participant. The blood sample was then taken from the donated blood bags into sterile test tubes, which were then used for closed-system screening. The collected blood was then centrifuged, the plasma separated, and the assays completed within 8 h. In the event that the analysis did not finish in 8 h, the samples stored at +2 ˚C to +8 ˚C for a maximum of 7 days and brought to room temperature before the analysis. The plasma sample was prepared by a senior blood bank laboratory worker using the fourth-generation Enzyme-Linked Immunosorbent Assay (ELISA) test kit (manufactured in Beijing Wantai Biological Pharmacy Enterprise Co., Ltd, Beijing, China) and analyzed for HBsAg.

2.7. Data quality assurance

The data collectors were selected based on their ability to speak the local languages, and researchers gave two days of guidance on the objective, data collection process, and field ethics. All data collections were supervised by the researchers. Each questionnaire was checked daily by the researchers for its integrity and consistency. The ELISA kits have been checked for proper storage conditions and expiration dates. Positive and negative internal controls were included in each test suite and standard operating procedures and manufacturer instructions were strictly followed. In a nearby area, a pre-test was carried out on 5% of the total sample to test the data collection tool, which included clarifications and corrections to the problems found in the questionnaire.

2.8. Data analysis

Each questionnaire was coded and manually checked for completeness, consistency, and clarity. The encoded and refined data was then entered into Epidata 3.1 and exported to SPSS version 20.0, where statistical analysis was performed. Descriptive analyzes were performed to determine the prevalence of HBV. A bivariate analysis was performed to select the candidate variables with $p<0.25$. Then multivariate analysis was carried out to identify independent predictors and control confounding factors. Multicolinearity was checked by using variance inflation factor and no problems were identified. Adequacy of the model was checked by Hosmer & Lemeshow goodness of test. Finally, independent variables at $p<0.05$ with an adjusted odds ratio (AOR) and its 95% confidence intervals (CI) were used in the multivariate logistic regression to declare statistical significance.

2.9. Operational definition

Knowledge of HBV prevention and control was measured using 18 questions, including general awareness of HBV, route of transmission, signs and symptoms, complications, and treatment according to the current literature. The correct answer was rated 1 and the wrong answer or "I do not know" was rated 0. Therefore, the knowledge score has been scaled from 0 to 18. A person with an overall score of 9 or less was classified as having poor knowledge, and those who scored 10 or more having good knowledge.

3. Results

3.1. Socio-demographic characteristics

A total of 548 voluntary blood donors took part in the study giving a response rate of 98.70%. The age of the respondents ranged from 18 to 51 years with a mean age of 24.84 years and a standard deviation (SD) of ± 5.85 years. Almost two-thirds of the participants (60.95%) were in the 18- 24 year-old age group and 330 (60.20%) were vocational school students (Table 1).

Table 1
Socio-demographic characteristics of volunteer blood donors in Jimma Zone, Ethiopia, 2018.

Characteristic	Categories	Frequency	Percent (%)
Age in years	18- 24	334	60.95
	25- 34	174	31.75
	≥35	40	7.30
Sex	Male	268	48.91
	Female	280	51.09
Marital status	Single	319	58.21
	Married	211	38.50
	Separated	18	3.29
Residence	Urban	444	81.02
	Rural	104	18.98
Educational level	Elementary	16	2.92
	Secondary	162	29.56
	College diploma	159	29.02
	Degree and above	211	38.50
Occupation	Government employed	130	23.72
	Self-employed	67	12.23
	Daily laborer/ jobless	21	3.83
	Student	330	60.22

3.2. Prevalence of Hepatitis B virus

Hepatitis B virus was detected by testing for HBsAg in serum. The overall prevalence of HBsAg among volunteer blood donors who participated in this study was 2.92%. The seropositivity of HBsAg in male participants was 12/268 (4.48%) which was higher than females which was 4/280 (1.43%). The age specific prevalence of HBsAg was highest (7.5%) among volunteer blood donors aged 35 years and over (Table 2).

3.3. Prevalence of HBV by clinical exposures and behavioral related factors

Regarding the distribution of HBsAg positivity by clinical exposure, 36 (6.57%) of the respondents had a history of hospitalization, and 12 (2.19%) of them had a history of blood transfusion. Most of the respondents 502(91.60%) had not been vaccinated against the HBV and had a higher prevalence of HBsAg 15 (2.4%) than those who were vaccinated, 46 (8.39%). The seropositivity of HBsAg among the hospitalized and blood transfused respondents was 2 (0.36%) and 1 (0.18%), respectively.

Although 23 (4.20%) of the study participants were exposed to injections of unsafe therapeutic drugs and their seropositivity for HBsAg was 3 (0.55%) higher than that of those who were not exposed, this was not statistically significant. Three hundred thirteen (57.12%) participants donated blood twice or more often. The prevalence of HBV infection was higher 11 (2.01%) among volunteer blood donors for the first time and blood donors with poor knowledge, 4 (0.73%) (Table 3).

3.4. Factors associated with HBV infection

Those candidate variables with $p<0.25$ bivariate logistic regression models were entered into multivariate logistic regression models in order to control confounding factors and to evaluate the net effects of risk factors on HBV infections and considered them as significant in the model when p-value was below 0.05.

Being male was an independent predictor of viral infection with hepatitis B. Male volunteer blood donors were approximately three times as positive for HBV infection as compared to female donors [AOR = 3.28, 95% CI: 1.01, 10.68].

The age group was also found as a risk factor for HBV infection. Blood donors in the age group 18–24 were 83% less likely to be seropositive for HBV infection than those in the age group ≥ 35 years [AOR = 0.17, 95% CI: 0.36, 0.78].

The number of blood donations was identified as a predictor of HBV infection. Voluntary blood donors who donated blood more than once

Table 2
Prevalence of HBsAg by socio-demographic characteristics of volunteer blood donors in Jimma Zone, Ethiopia, 2018.

Variable	Category	HBsAg Infection Status		COR (95%CI)	P-value
		Positive (%)	Negative (%)		
Age group (in years)	18- 24	5(0.91)	329(60.04)	0.19(0.04–0.82)	0.026*
	25- 34	8(1.46)	166(30.29)	0.59(0.15–2.35)	0.458
	≥ 35	3(0.55)	37(6.75)	1	
Sex	Male	12(2.19)	256(46.72)	3.23(1.03–10.16))	0.044*
	Female	4(0.73)	276(50.36)	1	
Marital status	Single	8(1.46)	311(56.75)	0.44(0.53–3.70)	0.448
	Married	7(1.28)	204(37.23)	0.58(0.068–5.02)	0.624
	Separated	1(0.18)	17(3.10)	1	
Residence	Urban	14(2.56)	430(78.47)	1.66(0.32–7.42)	0.507
	Rural	2(0.36)	102(18.61)	1	
Educational level	Elementary	0(0.00)	16(2.92)	0.000	0.999
	Secondary	5(0.91)	157(28.65)	0.93 (0.289–2.99)	0.640
	Diploma	4(0.73)	155(28.28)	0.75 (0.216–2.62)	0.654
	Degree and above	7(1.28)	204(37.23)	1	
Occupation	Employed	6(1.09)	124(22.63)	2.23 (0.74–6.77)	0.256
	Self employed	2(0.37)	65(11.86)	1.42(0.29–6.99)	0.666
	Daily laborer	1(0.18)	20(3.65)	2.31(0.271–19.68)	0.445
	Student	7(1.28)	323(58.94)	1	

Table 3
Prevalence of HBsAg by clinical and behavioral related factors among study participants in Jimma Zone, Ethiopia, 2018.

Variable	Category	HBsAg infection status		COR (95%CI)	P-value
		Positive (%)	Negative (%)		
Hospital admission	Yes	2(5.56)	34(94.44)	2.09(0.46–9.58)	0.342
	No	14(2.73)	498(97.27)	1	
Surgical procedure	Yes	1(4.35)	22(95.65)	1.55(0.19–12.23)	0.68
	No	15(2.86)	510(97.14)	1	
Blood transfusion	Yes	1(8.33)	11(91.67)	3.158(0.38–26.06)	0.286
	No	15(2.80)	521(97.20)	1	
Dental procedures	Yes	3(3.66)	79(96.34)	1.323(0.37–4.75)	0.667
	No	13(2.79)	453(97.21)	1	
Vaccinated against	Yes	1(2.17)	45(97.83)	0.721(0.09–5.59)	0.755
Hepatitis B virus	No	15(2.99)	487(97.01)	1	
Razor and sharp material	Yes	2(4.55)	42(95.45)	1.67(0.37–7.58)	0.509
sharing	No	14(2.78)	490(97.22)	1	
Unprotected multiple sex	Yes	1(5.26)	18(94.74)	1.89(0.24–15.12)	0.547
	No	15(2.84)	514(97.16)	1	
Tattooing, ear/nose	Yes	4(4.76)	80(95.24)	1.88(0.59–5.99)	0.283
pierce	No	12(2.59)	452(97.41)	1	
Unsafe therapeutic drug	Yes	3(13.04)	20(86.96)	5.91(1.56–22.39)	0.009*
injection	No	13(2.48)	512(97.52)	1	
Number of donations	Repeated	5(1.60)	308(98.40)	0.33(0.11–0.97)	0.043*
	First time	11(4.68)	224(95.32)	1	
Knowledge about HBV	Good knowledge	12(2.71)	431(97.29)	0.70 (0.22–2.23)	0.549
	Poor knowledge	4(3.81)	101(96.19)	1	

were 75% less likely to be seropositive for HBV infection than first-time donors [AOR = 0.25, 95% CI: 0.08, 0.76].

History of exposure to unsafe therapeutic drug injections was an independent predictor of HBV infection. Voluntary blood donors exposed to injections of unsafe therapeutic drugs were approximately seven times more likely to be seropositive for HBV infection than those who were not exposed [AOR = 6.98, 95% CI: 1.66, 29.29] (Table 4).

4. Discussion

In this study, the proportions of male and female voluntary blood donors were 48.91% and 51.09%, respectively, in contrast to previous studies from some parts of Ethiopia which showed that most blood donors were male donors; 87.9% in Gondar [21], 74.6% in Wolaita [21] and 98.7% in Jigjiga [21]. This could indicate that since most of the current study participants were university students, men and women became similarly aware of donating blood.

Most of the participants 334 (60.95%) in this study were between 18 and 24 years old. The results of this study partly coincided with a study in Hawasa 86.2% [21], in Arba Minch 80.2% [14], and with the WHO report that 45.0% of blood donors are 25 years or younger [11]. This indicated that the blood bank's routine activities were aimed at younger donors, mostly college and university students, as this group is perceived as more willing to donate blood and also as a low-risk group.

The prevalence of HBsAg among voluntary blood donors in the study area was 16 (2.92%). According to the WHO classification of HBV endemic, this is based on the HBsAg seropositivity rate, this result indicates that the study area is classified as a moderate endemic for HBV infection [4,6].

The prevalence of HBsAg in this study was higher than researches conducted in Eritrea with 2.58% and Egypt 2.30% [22,23]. This is likely due to differences in society, behavior, lifestyle, socioeconomic status, and levels of awareness. In contrast, it is lower than the study carried out in Yemen 5.10% [23], Cameroon 11.20% [24], and Nigeria 9.30% [23]. This could be because the majority of the study participants in the previous study were family substitutes and commercial blood donors, while this study only included voluntary blood donors.

The prevalence of HBV infection in this study was also lower than in similar studies in Ethiopia, 3.90% nationwide [25]; Bahir Dar 4.10% [26], Wolaita Sodo 9.60% [21] and Arba Minch 4.70% [14]. This could

Table 4
Factors associated with Hepatitis B viral infection among volunteer blood donors in Jimma Zone, Ethiopia, 2018.

Variable	Category	HBsAg Status		COR (95%CI)	AOR (95%CI)	P-value
		Positive	Negative			
Age group (years)	18–24	5	329	0.19(0.04–0.82)	0.17(0.36–0.78)	0.022*
	25–34	8	166	0.59(0.15–2.35)	0.53(0.13–2.24)	0.391
	≥ 35	3	37	1	1	
Sex	Male	12	256	3.19(1.02–1.01)	3.28(1.01–10.68)	0.049*
	Female	4	276	1	1	
Number of blood donation	Repeat	5	308	0.33(0.11–0.97)	0.25(0.08–0.76)	0.015*
	First time	11	224	1	1	
Unsafe therapeutic drug injection	Yes	3	20	5.91(1.56–2239)	6.98 (1.67–29.30)	0.080*
	No	13	512	1	1	

Keys: AOR: Adjusted Odds Ratio; COR: Crude Odds Ratio; CI: Confidence Intervals.

be due to previous studies focused on the type of replacement blood donation and an increased number of first-time donors. Some possible differences in the specificity and sensitivity of the screening tests used at different sites during the detection period could also partially explain the observed differences.

Being a male donor was about three times more likely to have HBV infection than a female donor. This finding was consistent with other studies previously conducted in other parts of Ethiopia [21,27]. This association might be attributed to cultural practices that could expose to HBV infection like circumcision with non-sterile equipment and in men's beauty salons that may use non-sterilize cleaning tools.

Participants aged 18–24 years-old were about 83% less likely to have HBV infection than those in the age group 35 years and above. This finding is in line with studies conducted at Jimma Blood Bank [24] and Gonder [21]. Similarly, a study in Nigeria shows that the prevalence of hepatitis B is highest in people over the age of 35 and has a statistically significant association with hepatitis B infection [23]. The 18–24-years-old age group is less exposed to risk factors such as occupational risk and unsafe sex.

Participants who had a previous history of blood donation were about 72% times less likely to be seropositive for HBV infection than those first-time donors. This finding is also in line with other studies conducted in Gonder where the prevalence of HBV among the first-time donor is about two times more likely to have HBsAg infection than repeated donors and Yemen where first-time donors had 22 times more likely to have HBsAg infection than the repeated donor [21,28]. This might be due to most new donors have not been previously tested for markers of infections that are used to exclude individuals from the donor population and repeated donors get awareness about the disease transmission.

Donors exposed to injections of unsafe therapeutic drugs were approximately seven times more likely to develop HBV infection than those who were not exposed. This study result was in agreement with studies carried out at Arba Minch [14] and the Jimma University Specialized Hospital [24]. According to WHO reports in developing countries; contaminated injections caused around 21 million HBV infections worldwide in 2000, which corresponds to 32% of all new infections [4]. This might be due to the therapeutic injection giver using contaminated syringes or other unsafe materials and procedures.

Most of the potential risk factors included in this study were not significantly associated with HBV infection. This could be due to the homogeneity of the study population, as most of the participants were young students.

5. Limitation of the study

The serum markers used to assess HBV infection were not complete; due to a lack of resources, HBV DNA was not detected by the polymerase chain reaction (PCR), which could increase the detection rate of HBV and thus its prevalence. Due to the requested earlier exposure history, a recall bias could be introduced.

6. Conclusions

The prevalence of HBsAg-positive HBV infection in volunteer blood donors in the study area was 2.92%, which is considered intermediate according to the WHO hepatitis B endemic category. The result of this study showed that low-risk blood donors were women, young people, repeat donors, and those who were not exposed to injections of unsafe therapeutic drugs. Therefore, the blood bank should raise awareness to repeat volunteer young donors and focus on donating blood among identified low-risk groups to keep blood safe for the blood donor-recipient and prevent HBV infection.

Data availability

All data supporting our findings are already described and included in the manuscript.

Declaration of Competing Interest

None.

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This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethical Review Committee of Jimma University, Institute of Health.

Informed consent statement

Written informed consent was obtained from each participant. Blood donors who tested positive for HBsAg during the study period were counseled by nurses assigned to the blood bank and linked to Jimma University Medical Center for further treatment. The Ethics committee approved that the finding of this study could publish since identifying images or clinical details of participants that compromise anonymity was not applicable.

Declaration of Competing Interest

The authors declare that they did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Debele Mekonnen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Tesfaye Solomon:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Mamo Nigatu:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

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